

Akhilesh Ayer

Indian students aspiring for academic careers often enter graduate programmes excited about “finally doing real research”, only to realise that it demands far more than they expected. Unlike undergraduate coursework, where projects come with predefined topics and structured guidelines, advanced research requires students to frame their own questions, design experiments, analyse unfamiliar datasets, think critically, navigate ethical protocols and communicate findings with clarity.

Difficult transition

This transition is especially difficult in a system where scientific temper, a value rooted in critical inquiry, evidence-based thinking and curiosity-led learning, is not cultivated early or consistently. Many students encounter real research expectations much later than their international peers, which makes the leap into rigorous academic work challenging. This raises a critical question: Are Indian students entering the global academic landscape without adequate preparation?

A 2023 survey found that undergraduate exposure to research remains significantly lower than postgraduate training. Another study among medical undergraduates revealed that only one in four had the opportunity to participate in practical



Rethinking research readiness

Are Indian universities preparing students for the world of academia?

research. The first meaningful experience with research began only during Master’s programmes, slowing skill development and denting confidence. One reason is the theory-heavy nature of many undergraduate programmes. Students learn scientific concepts but rarely engage in hands-on experimentation, data interpretation, or problem-solving. Practical work deepens understanding and builds abilities such as applying knowledge to real problems and transferring skills across disciplines.

Surveys also show that students value practical research exposure and want more opportunities. Wide variation Pockets of excellence are limited. India has more than 43,000 institutes offering undergraduate and postgraduate degrees but only a fraction are research-intensive or offer structured research opportunities for undergraduates. As a result, early research preparedness varies widely across the country. Even where training programmes exist, students

report the need for more comprehensive and up-to-date workshops. Thus, India has a vast pool of talented students but systemic gaps in early research exposure affect their academic progression and global competitiveness. India has long grappled with brain drain with limited opportunities, outdated infrastructure, and insufficient research funding contributing to this outflow. Countries like China show this trend can be reversed. Through initiatives such as Thousand Talents

Plan (TTP), scientists were provided with generous financial incentives, world-class research facilities, elite school options and a highly supportive environment, significantly boosting the country’s technological and research capabilities. India can also create conducive research environments that offer strong funding, modern infrastructure and long-term career incentives to retain and attract talent. International postgraduate programmes typically assume that undergraduates possess a baseline of

research readiness including experimental design, troubleshooting, academic writing, and familiarity with ethics. This puts many Indian students at a disadvantage, not because of lack of capability but because the education structure historically placed less emphasis on early research training. However, the landscape is changing. In recent years, institutions and policymakers have shown a clear shift towards strengthening research readiness from early on. Universities are adopting multidisciplinary and project-based curricula and offer undergraduate research internships that allow students to work under experienced scientists and engage with real-world problems. Research-intensive institutions are expanding short-term fellowships to give students hands-on experience. Policy changes At the policy level, the government has launched various reforms such as the Anusandhan National Research Foundation (NRF) and National Institutional Ranking Framework (NIRF) to prioritise high-quality research and academia-industry collaboration. The One Nation One Subscription (ONOS) initiative aims to widen access to scientific literature, ensuring students from diverse institutions can learn and explore ideas on par with global peers. There is growing emphasis on AI and digital literacy, recognising that modern re-

search requires familiarity with advanced tools and technologies. These developments show a growing recognition that research readiness and scientific temper should be foundational elements, not skills acquired later. With the world’s largest youth population and a rapidly expanding higher education system, sustained investment and long-term policy focus are essential. Equally critical is ensuring the effectiveness of these investments. For example, while ONOS brings a vast collection of subscription journals under one umbrella, the real impact will depend on how easily researchers can discover, navigate, and use the content. Priority should be given to embedding hands-on project opportunities in undergraduate curricula, upgrading laboratories and equipment, expanding updated training models and ensuring equitable access to global research resources. Equally important is cultivating scientific temper, research ethics and interdisciplinary problem solving capability from the earliest stages of higher education. Once India commits to these reforms, students will enter postgraduate programmes with greater confidence. More importantly, the nation will equip the next generation of researchers not only to participate in global science but to lead it.

The writer is the CEO of Cactus Communications

SCHOLARSHIPS

LEAP Emerge Fellowship Offered by the Leadership Education for Asian Pacifics (LEAP). Eligibility: Students in the final year of UG studies who are seeking an entry-level or internship position and demonstrate leadership experience through extracurricular activities and collaborative work. Rewards: \$3,000 stipend Application: Online Deadline: March 13 www.b4s.in/edge/LEAP1

Kiran Pratibha Scholarship Provided by the Kiran Foundation. Eligibility: Students who are at least 14 years old, have passed Class 8, have a monthly family income not exceeding ₹20,000 and are not receiving any other government scholarships. Rewards: Tuition fee plus benefits. Application: Online Deadline: April 30 www.b4s.in/edge/KRPS1

Nutanix NextGen Innovators Scholarship Offered by Nutanix. Eligibility: Indian students in the third or fourth year of UG programme in one of the specified disciplines. Rewards: \$2,500 Application: Online Deadline: May 31 www.b4s.in/edge/NINS1

Courtesy: Buddy4study.com



OFF THE EDGE Nandini Raman

I completed my Master’s in Computational Linguistics and am working as a content writer. What is the best path to be financially stable? Shwetha Rajeev

Dear Shwetha, A Master’s in Computational Linguistics is a hidden advantage that you are not monetising as yet as a content writer. You will need to pivot to working on making systems, models, strategy, and tech-language bridges. Some financially stable paths that fit your background are NLP/Language-AI roles if you have any coding exposure or are willing to build it. Target roles such as NLP engineer, language technologist, AI trainer or linguistic AI specialist, conversational AI designer, prompt engineer across product companies, AI startups, big tech vendors and Applied AI labs. Add skills like Python, NLP libraries (spaCy, NLTK, Hugging Face), basic ML concepts, LLM fine-tuning basics and evaluation of language models. Consider UX Writing and conversational design in roles like UX writer, content designer, conversational UX designer, voice assistant designer. You might need UX fundamentals, Figma basics, and conversation flows. You could also be a technical writer, API documentation specialist, AI product documentation for SaaS companies, developer tools and AI platforms. They pay well and need accuracy and domain understanding. AI content strategy or editorial ops such as content ops manager, AI content strategist, editorial programme manager will require SEO and analytics, workflow design, AI-assisted content systems and will be

Be prepared Uncertain about your career options? Low on self-confidence? This column may help

less hands-on writing and more management. I completed a B.Sc. (Hons) in Agriculture and want to pursue M.Sc. Agribusiness Management or other related courses. Which are the relevant programmes in India? Which colleges are good? Divya

Dear Divya, You can consider programmes such MBA or PGDM in Food and Agri-Business Management in IIM Ahmedabad or Lucknow, National Institute of Agricultural Extension Management (MANAGE), Hyderabad; Institute of Management Studies (BHU), or Symbiosis Institute of International Business (SIIB), Pune. Selection criteria include scores in entrance exams such as CAT, CMAT, SNAP and others, academic records, and interview. Many State Agricultural Universities offer M.Sc. or M.A. in Agricultural Economics and Business Management. You can also consider M.Sc. or M.Tech with specialisations related to Agribusiness for roles in agritech, analytics and agri-consulting. Other options could be M.Sc. or M.Tech in Agricultural Economics, Agri Business Economics, Agri Marketing or Farm Management or Rural Development and Food Processing or Agri-Food Engineering.

I completed M.A. English and qualified in the UGC-NET (JRF) and North-East State Level Eligibility Test. While pursuing Ph.D. admission, I am exploring career options beyond academia, particularly in creative

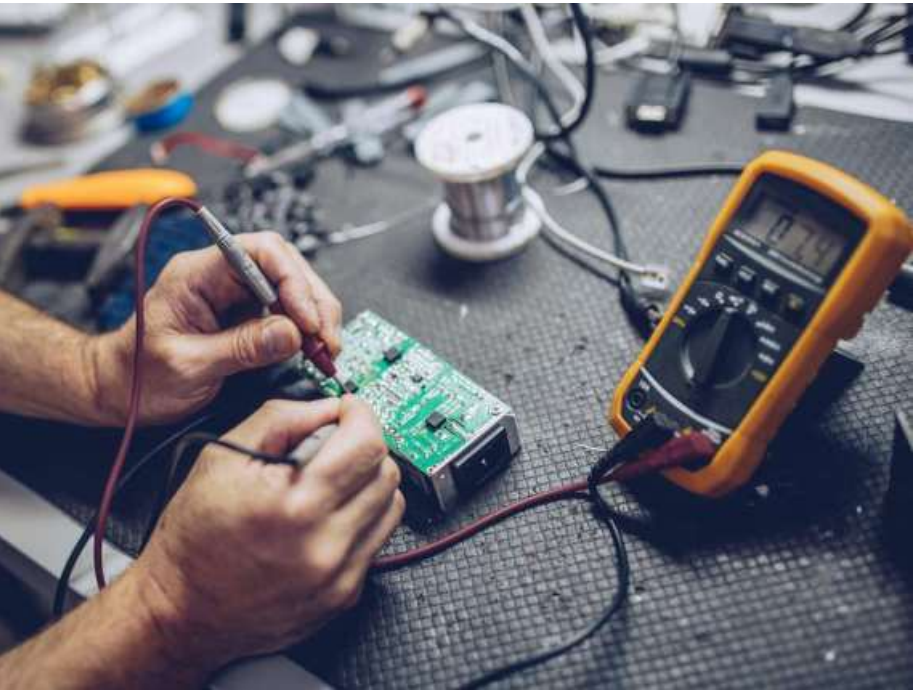
writing. How can I build a strong writing portfolio? Albina

Dear Albina, Alternative career paths beyond academia can involve being published in literary journals and magazines or scriptwriting for OTT or audio stories. Look for grants, fellowships, residencies and script and content labs. It will take time to establish your credibility. You can also consider roles such as editorial assistant or editor or manuscript evaluator across publishing houses, independent presses and literary agencies. Your JRF-level reading and critical skills are valuable. For financial stability and creative freedom, consider roles such as UX writer, content designer, or narrative designer for games and apps. You will need to upgrade skills in UX fundamentals, wireframe literacy and accessibility writing. You can also look for fellowships at think-tanks and writing labs or look for cultural documentation projects or language and heritage initiatives.

I am in Class 12 and considering a career in media and mass communication. What is the current scope? Which are the best institutions to look at? Layona

Dear Layona, Mass communication is about how information is created, distributed, and received across news media, advertising, public relations, digital platforms, film and broadcast, corporate communication, and more. It is both creative and strategic. One can do a

Bachelor’s course or work on specific diplomas/certificates and specialisations like digital media, advertising, PR, broadcast, film and TV. Careers include being a journalist, news anchor, copywriter, social media manager, content creator, digital marketer, radio/TV host and more. Today, there is a growing demand for digital content across digital media, social media platforms, podcasts, OTT streaming, brand storytelling, and online news. Many media professionals also work as freelancers, content creators, YouTubers, podcasters, and consultants. After Class 12 look for courses such as B.A. Journalism and Mass Communication, B.A. Mass Media, B.A. Mass Media and Communication or B.A. Multimedia and Journalism. Institutes to consider include Indian Institute of Mass Communication (IIMC), New Delhi; Symbiosis Centre for Media and Communication (SCMC), Pune; A.J.K. Mass Communication Research Centre (AJK MCRC), Jamia Millia Islamia, Delhi; Xavier Institute of Communications (XIC), Mumbai; Manipal Institute of Communication (MIC), Manipal; Indian Institute of Journalism and New Media (IIJNM), Bangalore; Department of Media Studies, Christ University, Bengaluru. Start doing your research and shortlisting the institutions. Look at their internship opportunities, industry connections, placements, faculty experience, alumni success stories and portfolio building support. Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers. The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge



Beyond books

How to transform academic libraries into innovation hubs

Koteswararao Anne

The word ‘library’ immediately brings images of an academic setting with long shelves of books, rows of desks, and an atmosphere of quiet concentration. Libraries are sanctuaries for reading, research and reflection. Despite being integral to campus life, they have seen little structural change, evolving primarily through the addition of new and specialised books. While this traditional role remains important, it no longer tells the whole story. With ever-growing demands of the market, students are not only expected to understand concepts but also to use them creatively and practically. This shift has pushed universities to rethink where and how learning happens.

Widening scope

An interesting development is the growing role of libraries. Apart from supporting the academic journey, they also support experiential learning. Along

with lending books, libraries have now begun lending equipment and technical tools. Unlike other major changes that require structure and process, the idea remains simple here. Libraries allow students to borrow electronic components, sensors, basic instruments or prototyping kits. This gives students the freedom to experiment, make mistakes and refine their ideas outside the scheduled laboratory hours. This approach is especially valuable in Engineering, Technology and Applied Sciences, where learning by doing is essential. Concepts that appear straightforward in lectures often reveal their complexity only when students attempt to implement them. When access to tools is limited to short lab sessions, experimentation can feel rushed and incomplete. By contrast, when students can take equipment with them, learning becomes more flexible and self-paced. They can test ideas

late at night, revisit designs multiple times and engage with the problem at hand. Accessibility Another important benefit of library-based equipment lending is inclusivity. Not all students can afford personal kits or devices. Even institutions cannot offer unlimited access to laboratories. However, libraries have always functioned on a simple and familiar system. They lend for a specific time frame and hold students accountable in case of loss. This can be extended to the lending of tools and equipment so that students no longer have to worry about spending money on multiple kits or devices to complete their coursework. Their ideas gain the space to develop, they can refine their experiments, try better techniques and understand how innovations work in real life. Libraries also become more than just a quiet study area. Students from different majors can connect share ideas. This is exactly how unique per-

spectives create tech-led innovations. Faculty members also stand to gain from this shift. Multiple courses require project-based or problem-based learning. At times, it is difficult to showcase the experiment explained in theory. With such an integration, teachers and professors can have readily available resources. Final-year projects as well as research initiatives reduce their dependency on departmental labs allowing teachers to focus on mentoring students, rather than solving logistical challenges. Deeper shift The National Education Policy 2020 stresses experiential learning, research orientation and skill development. With such a library, the criteria are met in the existing infrastructure. This model does not ask for dramatic shifts or expensive buildings. Libraries only have to continue what they have been doing: cataloguing, circulation, user training and accountability. With thoughtful planning, even modest collections of tools can make a meaningful difference. Ultimately, the changing role of the academic library reflects a deeper shift in education itself. Learning is no longer confined to absorbing information; it is about applying knowledge and solving problems. By opening their shelves to tools as well as books, libraries acknowledge this reality. Innovation in education does not always begin with grand announcements. Sometimes, it has to start quietly. When a student checks out a sensor instead of a textbook, he or she builds something with their own hands and discovers that learning can be as much about making as it is about reading. The writer is Dean, SVKM’s NMIMS Mukesh Patel School of Technology Management and Engineering (MPSTME).

